

Minimum Time 2D Car Optimal Control Problem (OCP)

An OCP aims to find a control function $u(t)$ that steers a dynamical system to a target state while minimizing a specific cost. For this project, we solve a **Minimum Time OCP** for a 2D vehicle.

Notations & Definitions

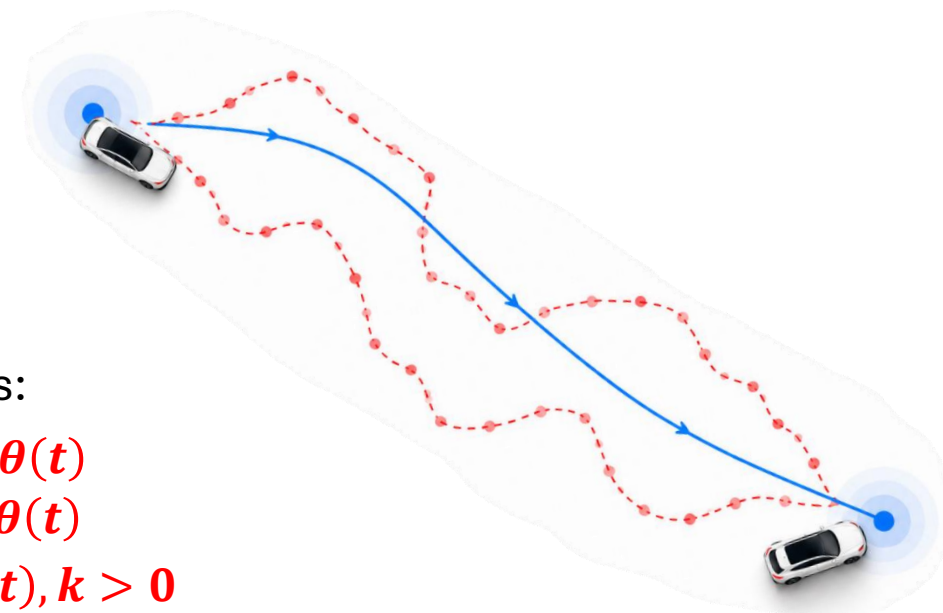
Before formulating the OCP, we define the vehicle parameters and variables :

- **State vector:** $\forall t \in [0, T], X(t) = (x(t) \ y(t) \ \theta(t))^T \in \mathbb{R}^3$
- **Control input:** $u(t) \in U \triangleq [-1, 1] \subset \mathbb{R}$
- **Initial state:** $X(0) = (0 \ 0 \ 0)^T$
- **Target state:** $X(T) = (x_1 \ y_1 \ \theta_1)^T \in \mathbb{R}^3$

Minimum Time OCP Formulation

Following the general OCP structure , the 2D car problem is defined as:

$$\min_{u \in L^\infty([0, T], U)} T \quad \text{subject to} \quad \begin{cases} \dot{x}(t) = \cos \theta(t) \\ \dot{y}(t) = \sin \theta(t) \\ \dot{\theta}(t) = ku(t), k > 0 \end{cases}$$



Standard OCP Cost Functional

$$J(u) = \underbrace{h(X(T))}_{\text{Terminal cost}} + \underbrace{\int_0^T f^0(t, X(t), u(t)) dt}_{\text{Running cost}}$$

$\Rightarrow$

Minimal Time Equivalence

$$J(u) = T = \int_0^T dt$$

$\Rightarrow$

$$\begin{cases} h(X(T)) = 0 \\ f^0(t, X(t), u(t)) = 1 \end{cases}$$

Applications : UAV (waypoint tracking), mobile robotics (warehouse navigation), autonomous parking...

Pontryagin Maximum Principle (PMP)

Hamiltonian

Let $\mathbf{p} = (p_x, p_y, p_\theta)^T \in \mathbb{R}^3$ the adjoint vector (representing the sensitivity of the cost relative to the state) and $p^0 \in \{-1, 0\}$ the cost multiplier.

$$H(t, X, u, \mathbf{p}, p^0) \triangleq p^T f(t, X, u) + p^0 f^0(t, X, u)$$

By substituting the specific 2D car dynamics and the running cost, we obtain the following Hamiltonian :

$$H(t, X, u, \mathbf{p}, p^0) = p_x \cos(\theta) + p_y \sin(\theta) + p_\theta k u + p^0$$

Adjoint system

Let $(\mathbf{u}^*, \mathbf{x}^*)$ be an optimal control-trajectory pair. There exist a transversality multiplier vector $\mathbf{p} \in \mathbb{R}^3$ and an adjoint $\mathbf{p}^*: [0, T] \rightarrow \mathbb{R}^3$ satisfying :

$$\begin{cases} \dot{\mathbf{p}}^*(t) = -\nabla_{\mathbf{x}} H(t, X^*, u^*, \mathbf{p}^*, p^0) \\ \mathbf{p}^*(T) = p^0 \nabla_{\mathbf{x}} h(X^*(T)) + \mathbf{p} J_{\mathbf{x}} g(X^*(T)) \end{cases}$$

After deriving the Hamiltonian, we obtain the following solutions

$$\begin{cases} \dot{p}_x(t) = 0 \\ \dot{p}_y(t) = 0 \\ \dot{p}_\theta(t) = p_x \sin(\theta) - p_y \cos(\theta) \end{cases}$$

Maximality conditions

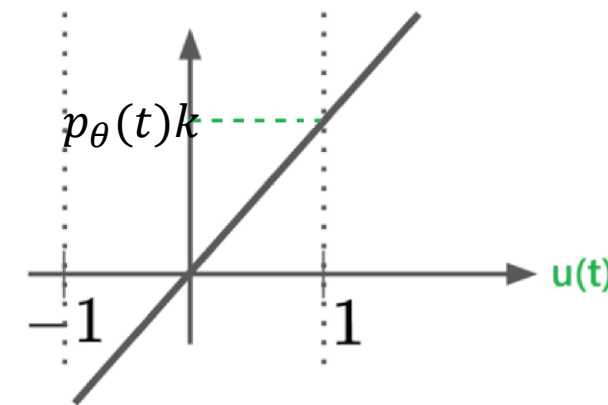
After solving $\mathbf{u}^*(t) \in \arg \max_{u \in [-1, 1]} H(t, X^*(t), u(t), \mathbf{p}^*(t), p^0(t))$, we obtain the optimal control :

$$\mathbf{u}^*(t) \in \arg \max_{u \in [-1, 1]} p_\theta(t) k u$$

Bang-bang proof

Assuming that $(p_x, p_y) \neq 0$, and knowing that $p_\theta(t)ku$ is a linear function, we are maximising a straight line. Thus, we obtain :

$$\begin{aligned} \text{Case 1 : } p_\theta(t) > 0 &\Rightarrow u^* = 1 \text{ (max value)} \\ \text{Case 2 : } p_\theta(t) < 0 &\Rightarrow u^* = -1 \text{ (min value)} \\ \text{Case 3 : } p_\theta(t) = 0 &\Rightarrow u^* = 0 \end{aligned} \Rightarrow u^*(t) = \begin{cases} 1 & \text{if } p_\theta(t) > 0 \\ -1 & \text{if } p_\theta(t) < 0 \\ 0 & \text{if } p_\theta(t) = 0 \end{cases}$$



Let us demonstrate that $(p_x, p_y) \neq 0$ by contradiction :

We assume that $(p_x, p_y) = (0,0)$ over $[0, T]$, this implies that $\dot{p}_\theta^*(t) = p_x \sin(\theta) - p_y \cos(\theta) = 0 \Rightarrow p_\theta^*(t) = C$.

But according to the PMP's final condition (if t_f is free), $H(t_f, X^*(t_f), u(t_f), p^*(t_f), p^0) = 0 \Leftrightarrow Cku^*(t) + p^0 = 0$

Case 1 : $p^0 = 0$

We obtain $C = 0 \Rightarrow p(t) = (0,0,0)$ and $p^0 = 0$,

⚠ Impossible regarding the non-triviality condition $((p^*(t), p^0) \neq 0 \text{ for all } t \in [0, t_f])$

Case 2 : $p^0 = -1$

We obtain $Cku^*(t) = 1 \Rightarrow |C| = \frac{1}{k} \Rightarrow C > 0$ over $[0, t_f]$, or $C < 0$ over $[0, t_f]$

⚠ Impossible, the car is physically trapped in an infinite circle, making a generic target unreachable

$\Rightarrow$ By contradiction, we demonstrated that $(p_x, p_y) \neq 0$

Numerical method to solve OCP

Problem statement

$$\begin{aligned}\dot{x} &= \cos(\theta), & \dot{y} &= \sin(\theta), \\ \dot{\theta} &= ku, & u &\in [-1; 1]\end{aligned}$$

Key Python libraries

- Numpy
- scipy.optimize.minimize
- scipy.optimize.fsolve
- matplotlib

Discretization

$$\begin{aligned}x_{i+1} &= x_i + \cos(\theta_i) \Delta t, \\ y_{i+1} &= y_i + \sin(\theta_i) \Delta t \\ \theta_{i+1} &= \theta_i + ku_i \Delta t\end{aligned}$$

Cost functions

$$\begin{aligned}J_2 &= -\sum_{i=0}^{N-2} u_i^2 \Delta t \\ J_3 &= T - \varepsilon \sum_{i=0}^{N-2} u_i^2 \Delta t\end{aligned}$$

Step 1 : Smooth BSB

The result is **Success : False**

- The approximation cannot satisfy the terminal constraints.
- It captures the correct BSB.

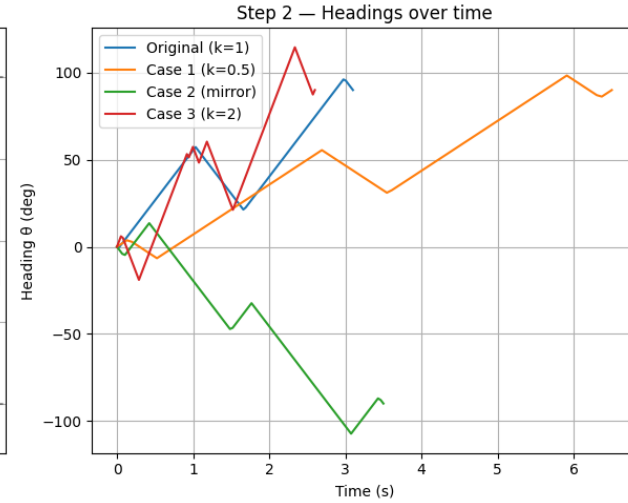
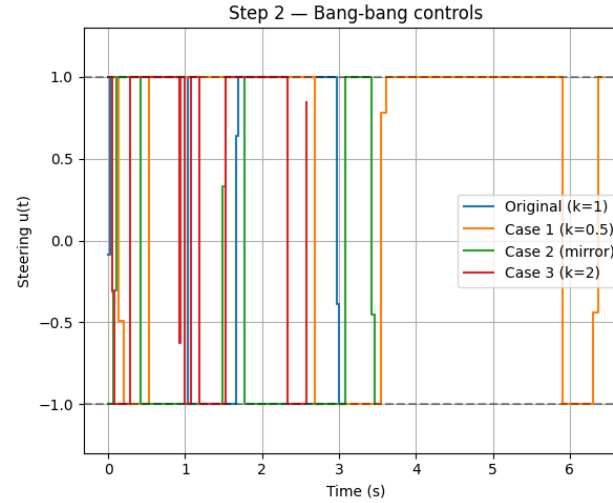
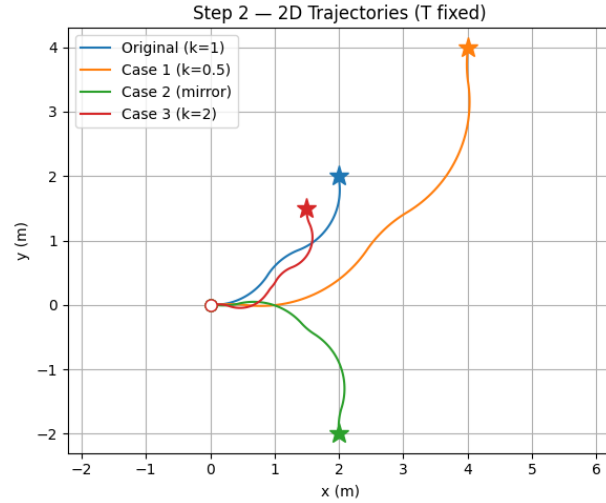
Step 2 : Full trajectory optimization at fixed T

- Interpolated to N = 100 grid points
- The horizon T is fixed slightly above the expected optimum
- The result is a nearly perfect bang-bang control

Step 3 : Free final time & exact BSB refinement

- Decreasing tolerances $10^{-3} \rightarrow 10^{-4} \rightarrow 10^{-5} \rightarrow 10^{-6}$
- Find the exact switching times t_1 and t_2 of the BSB control
- The residual integrates the exact bang-bang dynamics forward in time

Results & Interpretations



Additional study : Cart with viscous friction

(train stopping at a station, spacecraft rendezvous, autonomous vehicle parking)

$$\dot{X}(t) = (\dot{x} \ \dot{v})^T = (v(t) \ u(t) - \beta v(t))^T \in \mathbb{R}^2,$$

where $\beta > 0$ is the friction

System dynamics

$$u(t) \in U \triangleq [-1, 1] \subset \mathbb{R}$$

Control input

$$OCP \begin{cases} \min_{u \in L^\infty([0,T],U)} T \\ \dot{X}(t) = f(X(t), u(t)), t \in [0, T] \\ X(0) = (0 \ 0)^T, X(T) = (d \ 0)^T \end{cases}$$

$$H(t, X, u, p, p^0) = p_x v + p_v (u - \beta v) + p^0$$

Hamiltonian

Adjoint system

$$\begin{cases} \dot{p}_x^*(t) = 0 \\ \dot{p}_v^*(t) = \beta p_v^*(t) - p_x \end{cases}$$

Maximality condition

$$u^*(t) \in \arg \max_{u \in [-1;1]} p_v^*(t) u$$

